Touchless Touch screen. It consists of an LCD TV screen which is laden with sensors that detect hand movements. By merely waving the hands in front of the screen, you can interact with what is being displayed. For example, you could flip through pages by making a page-flipping motion, or gesture left or right to move through objects. Interactive displays such as these are likely to find use as advertisement kiosks, and might one day make it to our living rooms and meeting rooms.

3D TV

3D technologies promise to take television into the three-dimensions that we are accustomed to in the real-world! There are two possible implementations here—both actively researched, one more likely to visit our sets sooner rather than later. Let's take a look at the more fantastical possibility.

Brewing in the research labs of Mitsubishi Electric is an actual 3D TV, a "complete end-to-end 3D TV system that performs real-time acquisition, transmission, and 3D display of dynamic scenes." What the 3D TV system consists of is an array of cameras which is synchronized and together captures different views of a scene. To deal with the high processing and bandwidth requirements, the system uses a fully distributed architecture with clusters of PCs. This forms the 'acquisition' side of the 3D TV. To output the captured video, the system uses a multi-projector display with horizontal parallax. Since we are talking actual 3D here, the researchers also had to implement anti-aliasing techniques to improve the rendering of images on the 3D display. But the result is astounding in its promise—a true 3D scene on our television screens, hopefully, not too far from now!

Slightly less exciting, but cool still, is a 3D-like display; not actual 3D but good enough! The prominent technology here is Philips' so-called WOWvx—implemented and showcased by television manufacturers such as LG. WOWvx fools our eyes into true stereoscopic vision and depth perception—two important ways in which our eyes perceive a 3D environment. The tech consists of a series of tiny cylindrical lenses which split light from a TV into multiple parts and scatter them to multiple directions. This helps our brain perceive the image as three-dimensional. The technology also adds a "depth" map to the visual output to give our brains the sense of depth in the image. Some prototypes based on WOWvx technology can switch between 2D and 3D on the fly.

WOWvx brings depth perception to our flat television screens

Volumetric Displays

Holographic projectors: R2-D2 showing Obi-Wan a message from Princess Leia, could that be in our future? Volumetric displays are an attempt to break out of the 2-dimensional confines of the television frame. The most successful of volumetric displays is perhaps the Heliodisplay, which although isn't really from this category, but is close enough for us to consider here. The Heliodisplay actually works on a projection screen; in its case the screen is made up of a thin film of water—mist, literally. The water is sprayed into the general atmosphere and then images and video are projected on to this film. Some variations of this commercial product come with touch support, allowing you to manipulate objects that hang in mid air.

Then we have the displays which rely on persistence of vision to create a volumetric object in space. With these kinds of "swept-surface" displays, an object to be viewed is generally cut into cross-sectional slices which are then projected one after another, and quickly to form the entire object. For example, one implementation sees fast-moving LEDs create a 360-degree surface in air. Another implementation might see a pulsing laser—each pulse lasting about a nanosecond—creating globes of plasma in air. These superheated plasma spheres are then combined to form an object.

While there might be little content for our HDTVs, things are certainly looking up from last year. Prices are falling and will only go down, terrestrial TV is moving towards HD, slowly, and more excitingly, there is a bevy of cool tech just on the horizon to keep us wondering and eagerly looking forward to the coming years. ■

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